

INSTRUCTIONS

You have 2 hours and 50 minutes to complete the exam.

- The exam is closed book, closed notes, closed computer/calculator, except the provided final reference sheet.
- Mark your answers on the exam itself in the spaces provided. We will not grade answers written on scratch paper or outside the designated answer spaces.
- If you need to use the restroom, bring your phone and exam to the front of the room.

For questions with **circular bubbles**, you should fill in exactly *one* choice.

- ☐ You must choose either this option
- ☐ Or this one, but not both!

For questions with **square checkboxes**, you may fill in *multiple* choices.

- ☐ You could select this choice.
- ☐ You could select this one too!

****Important****: Please **fill in** circles and squares to indicate answers and cross out or erase mistakes.

Preliminaries

You can complete these questions before the exam starts.

- (a) What is your full name?

- (b) What is your student ID number?

- (c) Who is sitting to your left? (Write *no one* if no one is next to you.)

- (d) Who is sitting to your right? (Write *no one* if no one is next to you.)

1. (28.0 points) True or False

- (a) (2.0 pt) The height of each bar in a histogram represents the proportion of data within the corresponding bin.
- ☐ True
- ☒ False
- (b) (2.0 pt) According to the Case Study lecture, the pixel of an image is typically considered a categorical variable when building machine learning models.
- ☐ True
- ☒ False
- (c) (2.0 pt) A classifier is considered to be overfitting if it performs very well on the test set.
- ☐ True
- ☒ False
- (d) (2.0 pt) If you are a subject in an experiment, knowing whether you are in the treatment or control group can be considered a confounding variable.
- ☒ True
- ☐ False
- (e) (2.0 pt) For any distribution, the percentage of data that lies beyond two standard deviations on either side of the mean is less than 30%.
- ☒ True
- ☐ False
- (f) (2.0 pt) For any regression line, the SD of the residuals is equal to the root mean squared error.
- ☒ True
- ☐ False
- (g) (2.0 pt) We sample a table with replacement to shuffle labels for A/B testing.
- ☐ True
- ☒ False
- (h) (2.0 pt) According to the Central Limit Theorem, if a sample is drawn at random from the population with replacement, then the probability distribution of the sample average is normal, regardless of the sample size.
- ☐ True
- ☒ False
- (i) (2.0 pt) DJ Patil covered climate change in depth during his guest lecture.
- ☐ True
- ☒ False

(j) (2.0 pt) The median of a set of 15 integers will always be an integer.

- ☒ True
☐ False

(k) (2.0 pt) Suppose a hypothesis test is proposed and we already know that the null hypothesis is true. If 100 researchers each independently collect a large random sample of the same size to carry out an experiment and they all use 5% as their p-value cutoff, we should expect around 95% of them to fail to reject the null.

- ☒ True
☐ False

(l) (2.0 pt) Suppose you fit a regression line to two data sets: (A) the original data set; and (B) the dataset with a few outliers (with respect to the y-axis) removed. The line for (A) will have a larger slope (in absolute value) than the line for (B).

Assume both data sets are standardized just prior to fitting the lines.

- ☒ True
☐ False

(m) (2.0 pt) If we use linear regression to predict y -values based on our x -values, the median of our residuals will always be zero.

- ☐ True
☒ False

(n) (2.0 pt) Given a function `error(a, b)` which computes some error based on its input arguments, a valid output from the call `minimize(error)` could be an array containing two elements: 4 and 20.

- ☒ True
☐ False

2. (32.0 points) Hawaii

Armond and Belinda are studying the characteristics of the guests who stay at their hotel in Hawaii.

To get a sense of this, they plan to randomly sample guest bookings from the hotel's database.

(a) (14.0 points)

Their initial project is to analyze guest spend, the amount spent per guest booking in dollars.

- i. **(2.0 pt)** Suppose Armond wants to randomly sample guest bookings to create a **68%** confidence interval for the **population mean** of guest spend, and he knows that the population SD is 500 dollars.

What is the minimum sample size he needs to create a confidence interval that has a width of 50 dollars?

- ☐ 2500
- ☐ 1600
- ☐ 800
- ☒ 400
- ☐ 100
- ☐ 20
- ☐ 10
- ☐ There is not enough information to answer

- ii. **(2.0 pt)** Suppose Armond wants to randomly sample guest bookings to create a **95%** confidence interval for the **population median** of guest spend, and he knows that the population SD is 500 dollars.

What is the minimum sample size he needs to create a confidence interval that has a width of 40 dollars?

- ☐ 2500
- ☐ 1600
- ☐ 800
- ☐ 400
- ☐ 100
- ☐ 20
- ☐ 10
- ☒ There is not enough information to answer

- iii. (3.0 pt) Suppose that Belinda creates her own random sample of 100 guest bookings. She observes a sample average of 4000 dollars for guest spend and she also knows that the population SD is 500 dollars.

What is her **99.7%** confidence interval for the true **population mean** of guest spend (in dollars)?

- ☐ (3990, 4010)
☐ (3985, 4015)
☐ (3950, 4050)
☐ (3900, 4100)
☒ (3850, 4150)
☐ (3500, 4500)
☐ (3000, 5000)
☐ There is not enough information to answer

- iv. (4.0 pt) Armond suspects that the spend at the Hawaii hotel is lower than the spend at their sister hotel in Sicily, which has an average guest spend of 7000 dollars across all of its bookings.

Based on a **95%** confidence interval for the **population mean** guest spend time of (5500, 7500) dollars, if his p-value cutoff is **5%**, what should he conclude?

Select all that apply.

- ☒ The data are consistent with the hypothesis that the distribution of guest spend is the same for both the Hawaii hotel and the Sicily hotel.
☐ The data are consistent with the hypothesis that the guest spend is higher on average at the Sicily hotel than at the Hawaii hotel.
☐ The data are consistent with the hypothesis that the guest spend is higher on average at the Hawaii hotel than at the Sicily hotel.
☐ There is not enough information to make any of these conclusions.

- v. (3.0 pt) Suppose that Belinda creates ten different random samples, each of 100 guest bookings, and constructs a **90%** confidence interval for the true **population mean** of guest spend using each sample. What is the chance that at least one of the intervals contains the population mean?

$$1 - (0.10)^{10}$$

(b) (4.0 points)

Note: This section does not depend on the previous parts.

Armond thinks it could be a good idea to predict guest spend from how far away they live from the hotel.

He collects a random sample of guest bookings and creates a table with two columns:

- *Spend*: (float) the amount spent (in dollars) by a guest during their stay
- *Distance*: (float) the number of miles the guest traveled to get to the hotel

Suppose that:

- the '**Spend**' column has a mean of 5,000 and a standard deviation of 500
- the '**Distance**' column has a mean of 4,000 and a standard deviation of 1,000
- the correlation between the '**Spend**' and '**Distance**' columns is 0.8

- i. **(2.0 pt)** Armonds fits a regression line using the table above to predict '**Spend**' from '**Distance**'.

What is the **intercept** of this regression line?

- ☐ 21,000
- ☐ 12,000
- ☐ 6,600
- ☒ 3,400
- ☐ -4,000
- ☐ -11,000
- ☐ None of the above

- ii. **(2.0 pt)** For a guest from Berkeley, who would have to travel 2,000 miles to get to Hawaii, what would this regression line predict as their spend?

- ☐ 5,800
- ☒ 4,200
- ☐ 4,000
- ☐ 2,500
- ☐ 2,000
- ☐ None of the above

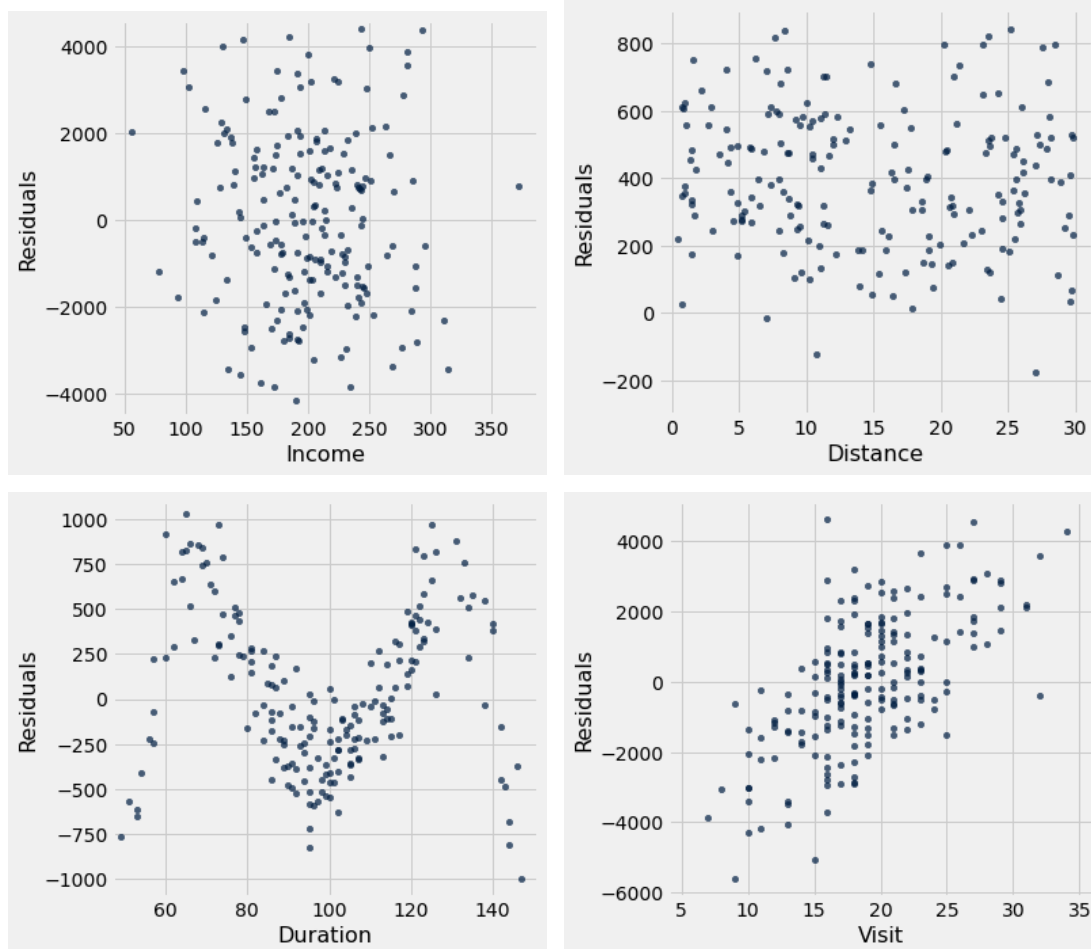
(c) (6.0 points)

Note: This section does not depend on the previous parts.

Belinda decides to predict the guest spend (in dollars) from each of the following four different variables using least squares linear regression:

- *Income*: (float) the guest's household income (in thousands of USD)
- *Distance*: (float) the distance (in thousands of miles) of the guest's home from the hotel
- *Duration*: (int) the number of days the guest stayed at the hotel for that booking
- *Visit*: (int) the guest's visit number (e.g. a value of 10 implies the booking was the guest's 10th visit to the hotel)

To assess her predictions from each variable, she creates the following residual plots:



i. (2.0 pt) Which of the plots above indicate that the variable is linearly associated with guest spend?

Select all that apply.

- ☒ Income
- ☐ Distance
- ☐ Duration
- ☐ Visit
- ☐ None of the above.

- ii. (2.0 pt) Belinda suspects she may have made a mistake in her code when plotting the residuals. Which of the plots above are impossible residual plots?

Select all that apply.

- ☐ Income
- ☒ Distance
- ☐ Duration
- ☒ Visit
- ☐ None of the above.

- iii. (2.0 pt) For which of the above plots should Belinda try a nonlinear equation for regression?

Select all that apply.

- ☐ Income
- ☐ Distance
- ☒ Duration
- ☐ Visit
- ☐ None of the above.

(d) (8.0 points)

Note: This section does not depend on the previous parts.

Belinda notices that guests' nationality seems to differ quite a bit depending on whether they request a late check out.

She creates a table with the following columns:

- 'Nationality': (string) the country where the guests reside (e.g. 'Italy', 'France')
- 'Late Checkout': (string) whether the guests requested a late check out ('Yes' or 'No').

She decides to conduct a hypothesis test to determine whether the observed difference between guests who request a late checkout and those who do not is due to chance.

i. (2.0 pt) Which of the following is a null hypothesis that Belinda could use for her test?

Select all that apply.

- ☐ The 'Late Checkout' distribution is the same among guests of all nationalities.
- ☐ The 'Late Checkout' distribution is more varied among domestic guests compared to international guests.
- ☒ The 'Nationality' distribution is the same between guests who request a late checkout and guest who do not.
- ☐ The 'Nationality' is more diverse among guests who request a late checkout compared to guest who do not.
- ☐ The 'Nationality' distribution is different between guests who request a late checkout and guest who do not.
- ☐ None of the above

ii. (2.0 pt) Which of the following could be used as a test statistic for this hypothesis test?

Select all that apply.

- ☐ The difference in standard deviation of 'Nationality' between guests who request a late checkout and guests who do not.
- ☐ The total variation distance between the 'Late Checkout' distributions of domestic guests and international guests.
- ☐ The difference of means in 'Nationality' between guests who request a late checkout and guest who do not.
- ☒ The total variation distance between the 'Nationality' distributions of guests who request a late checkout and guest who do not.
- ☐ The absolute difference of means in 'Nationality' between guests who request a late checkout and guests who do not.
- ☐ None of the above

- iii. (2.0 pt) Which of the following approaches could be used to create simulations under the null hypothesis?

Select all that apply.

- ☐ Sampling all of the rows of the table without replacement
- ☒ Shuffling just the 'Nationality' column
- ☒ Shuffling just the 'Late Checkout' column
- ☐ Calling `sample_proportions` and passing in the observed 'Nationality' distribution as an argument
- ☐ Calling `sample_proportions` and passing in the observed 'Late Checkout' distribution as an argument
- ☐ None of the above

- iv. (2.0 pt) Suppose the 5th percentile of the simulated test statistics is 0.1 and Belinda is using a 5% cutoff. Which of the following conclusions can she make?

- ☐ The data are consistent with the null hypothesis.
- ☐ The data are consistent with the alternative hypothesis.
- ☐ We cannot make a conclusion because we do not what the alternative hypothesis is.
- ☐ We cannot make a conclusion because we do not what the observed test statistic is.
- ☒ None of the above

3. (18.0 points) Will Frank Be on Time?

In the year 2031, Rebecca and Sarah are excited to see Frank Ocean perform at a festival.

Based on historical data, they know there's a 30% chance he will be on time, a 60% chance he will be late, and a 10% chance he doesn't show up at all.

Thankfully, they have found a website, *willfrankbeontime.com*, that attempts to predict the outcome of Frank Ocean's arrival. It returns one of the following:

- 'yes', if Frank will show up on time
- 'no', if Frank will show up late or will not show up at all

If he is on time, the website returns 'yes' 85% of the time. If he is late, the website returns 'yes' 11% of the time. If he does not show up, the website returns 'yes' 8% of the time.

Hint: it will help to draw a tree diagram.

- (a) (2.0 pt) What is the probability that Frank will be on time *or* show up late?

Please leave your answer as an expression.

$0.3 + 0.6$

- (b) (2.0 pt) If we define “negative” as Frank being late or not showing up, and we define “positive” as Frank being on time, what is the false negative rate of the website?

Please leave your answer as an expression.

0.15

- (c) (2.0 pt) What is the probability that Frank is on time and the website returns 'no'?

Please leave your answer as an expression.

$0.3 * 0.15$

- (d) (3.0 pt) Suppose the website returns “yes”. Which of the following represents the probability that Frank was on time, given that the website returned 'yes'?

- ☐ 0.3×0.85
☐ $0.3 \times 0.85 \times 0.11 \times 0.08$
☒ $\frac{0.3 \times 0.85}{0.3 \times 0.85 + 0.6 \times 0.11 + 0.1 \times 0.08}$
☐ 0.85
☐ $\frac{0.85 \times 0.11 \times 0.08}{0.3 \times 0.85 + 0.6 \times 0.11 + 0.1 \times 0.08}$

- (e) (3.0 pt) Suppose Frank shows up late. What is the probability that the website correctly predicted that he was not on time?

- ☐ 0.6×0.89
☐ $0.15 \times 0.89 \times 0.92$
☐ $\frac{0.6 \times 0.89}{0.3 \times 0.15 + 0.6 \times 0.89 + 0.1 \times 0.92}$
☒ 0.89
☐ $\frac{0.15 \times 0.89 \times 0.92}{0.3 \times 0.15 + 0.6 \times 0.89 + 0.1 \times 0.92}$

- (f) (3.0 pt) What is the probability that the website returns 'yes' and Frank Ocean is late or does not show up?

- ☒ $0.6 \times 0.11 + 0.10 \times 0.08$
☐ $1 - \frac{0.3 \times 0.85}{0.3 \times 0.85 + 0.6 \times 0.11 + 0.1 \times 0.08}$
☐ $(0.6 + 0.10) \times 0.11$
☐ $\frac{0.6 \times 0.11 + 0.1 \times 0.08}{0.3 \times 0.85 + 0.6 \times 0.11 + 0.1 \times 0.08}$
☐ $\frac{0.11 + 0.08}{2}$

- (g) (3.0 pt) Suppose that the morning of a performance, the weather forecast isn't looking too great, so Sarah & Rebecca think that there's a 20% chance Frank is on time, 35% chance he is late, and 45% chance he does not show up.

Suppose Sarah & Rebecca then consult the website and it returns 'no'.

Given this new information, what is Sarah & Rebecca's subjective probability that Frank does not show up?

- ☐ 0.45
☐ $1 - (0.2 \times 0.35)$
☐ 0.45×0.08
☒ $\frac{0.45 \times 0.92}{0.2 \times 0.15 + 0.35 \times 0.89 + 0.45 \times 0.92}$
☐ $\frac{0.15 \times 0.92}{0.25 \times 0.15 + 0.6 \times 0.89 + 0.15 \times 0.92}$

4. (38.0 points) To Buy or Not to Buy?

Siobhan and Roman are trying to understand the performance of Waystar, a media organization that owns multiple businesses.

They put together a table called **performance**, which contains randomly sampled public information about the performance of Waystar's businesses over the last 40 years. Here are the first few rows:

Name	Year	Revenue	Profit	Buy
Vaultier	2022	120.9	70.2	1
Adventure Parks	2020	59.2	25.8	0
ATN	2021	71.7	58.7	1
Brightstar Cruises	2020	42.3	28.9	0
Adventure Parks	2019	52.8	26.3	0

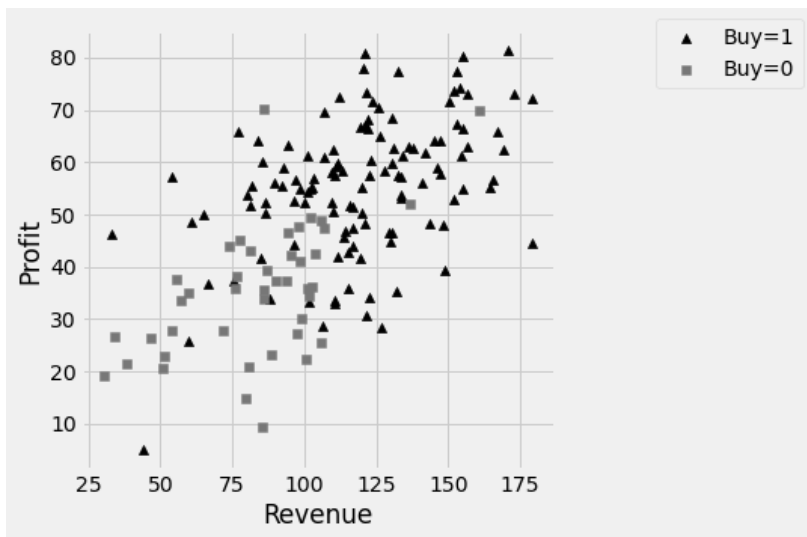
... (158 rows omitted)

The table contains the following columns:

- *Name*: (string) the name of the business
- *Year*: (int) the year of the financial performance
- *Revenue*: (float) the business's revenue (in millions of USD)
- *Profit*: (float) the business's profit (in millions of USD)
- *Buy*: (int) whether Wall Street analysts recommend buying the business's stock (1 is "yes", 0 is "no")

(a) (10.0 points)

Roman uses the table to make the following scatterplot in an attempt to classify whether a business will receive a 'Buy' recommendation from Wall Street analysts.



i. (2.0 pt) Suppose that in 2023, a Waystar business has Revenue of 75 and Profit of 70.

If Roman uses a 3-nearest neighbor classifier, what will the prediction be?

- ☒ 1
☐ 0
☐ Not enough information to tell.

- ii. (2.0 pt) Suppose that in 2023, a Waystar business has Revenue of 70 and Profit of 45.

If Roman uses a 5-nearest neighbor classifier, what will the prediction be?

- ☐ 1
☒ 0
☐ Not enough information to tell.

- iii. (2.0 pt) Suppose that instead of using a k-nearest neighbor classifier, Roman wants to use a d -distance classifier.

With this classifier, instead of considering the k closest neighbors, all neighbors within a distance d are considered.

Suppose that in 2023, a Waystar business has Revenue of 55 and Profit of 43.

If Roman decides to try $d = 10$, what will the prediction be?

- ☐ 1
☒ 0
☐ Not enough information to tell.

- iv. (4.0 pt) Suppose Siobhan wants to find the best possible value of d that will minimize the prediction error of Roman's d -distance classifier.

She creates the following names her code:

- `train`: a table that contains a random sample of 100 rows from `performance`
- `test`: a table that contains the rows from `performance` that are not in `train`
- `classify`: a function that takes in `train`, `test`, and `d` (the distance considered by the classifier), and returns a prediction for the example's 'Buy' recommendation

She then writes the following code:

```
def error(d):
    num_errors = 0
    for example in test.rows:
        num_errors = num_errors + (classify(train, example, d) == example.column('Buy'))
    return num_errors / test.num_rows
```

```
best_d = minimize(error)
```

Which of the following statements will occur when the above code is run?

Select all that apply.

- ☐ `best_d` will contain the value of d that provides the highest accuracy on the test set
☐ `best_d` will contain the value of d that provides the highest accuracy on the training set
☐ The last line of code will result in an error because d must be an integer
☐ `best_d` will contain the minimum possible error that can be achieved on the test set
☐ `best_d` will contain the minimum possible error that can be achieved on the training set
☒ None of the above.

Clarification: option 1 and option 6 were both accepted as correct due to a typo.

(b) (15.0 points)

Note: This section does not depend on the previous parts.

Suppose that instead of classifying whether a business will get a 'Buy' recommendation, Roman would like to use nearest neighbors to predict a business's *probability* of getting a 'Buy' recommendation.

We will use the following method:

- i. Find the k nearest neighbors of the business with respect to 'Revenue' and 'Profit'.
- ii. For each of the k nearest neighbors, define their **weight** as follows. Let d_r be the difference between the neighbor's revenue and the unclassified businesses's revenue. Let d_p be the difference between the neighbor's profit and the unclassified businesses's profit. Calculate the neighbor's weight as

$$\text{weight} = \frac{1}{1 + \sqrt{d_r^2 + d_p^2}}$$

- i. Among the k nearest neighbors, find the weighted proportion that received a 'Buy' rating. For example, if $k = 3$ and the 3 nearest neighbors have 'Buy' values of 0, 0, & 1 with weights 0.9, 0.3, & 0.2, respectively, then the weighted proportion that received a 'Buy' recommendation is

$$\frac{\overbrace{0.9}^{\text{weight}} * \overbrace{0}^{\text{Buy}} + \overbrace{0.3}^{\text{weight}} * \overbrace{0}^{\text{Buy}} + \overbrace{0.2}^{\text{weight}} * \overbrace{1}^{\text{Buy}}}{\underbrace{0.9 + 0.3 + 0.2}_{\text{sum of weights}}}$$

In order to estimate the 'Buy' probability of a new business using the scheme above, Roman defines a function `knn_prob` that takes the following arguments:

- *train*: A three-column table in which the first column is labeled 'Revenue', the second column is labeled 'Profit', and the third column is labeled 'Buy'. Each row of the table represents a business in the training set.
- *business*: An array of length two containing the business revenue and profit (in that order) of the business to predict on.
- *k*: The value of k to use for k -nearest-neighbors.

The function returns the weighted proportion that received a 'Buy' recommendation among the k -nearest-neighbors, according to the scheme proposed above. Here is the function, partially completed:

```
def knn_prob(train, business, k):

    revenue_diffs = train.column('Revenue') - _____
                                                    (a)

    profit_diffs = train.column('Profit') - _____
                                                    (b)

    train_dist = train.with_column('Distance', _____)
                                                    (c)

    nn = train_dist.sort(_____.).take(_____.)
                        (d)                (e)

    weights = 1/(1 + _____)
                    (f)

    return np.sum(_____.) / np.sum(weights)
```

(g)

i. (2.0 pt) Fill in blank (a).

```
business.item(0)
```

ii. (2.0 pt) Fill in blank (b).

```
business.item(1)
```

iii. (3.0 pt) Fill in blank (c).

```
np.sqrt(revenue_diffs ** 2 + profit_diffs ** 2)
```

iv. (2.0 pt) Fill in blank (d).

```
'Distance'
```

v. (2.0 pt) Fill in blank (e).

```
np.arange(k)
```

vi. (2.0 pt) Fill in blank (f).

```
nn.column("Distance")
```

vii. (2.0 pt) Fill in blank (g).

```
nn.column("Buy") * weights
```


(c) (13.0 points)

Note: You may use the `knn_prob` function from the previous question for this section.

Suppose Siobhan wants to understand the variability in Roman's nearest neighbor regression predictions.

She wants to construct a confidence interval for the probability of getting a 'Buy' recommendation by bootstrapping the sample data.

Siobhan defines a function `prob_interval` that takes the following arguments:

- *train*: A three-column table in which the first column is labeled 'Revenue', the second column is labeled 'Profit', and the third column is labeled Buy. Each row of the table represents a business in the training set.
- *business*: An array of length two containing the business revenue and profit (in that order) of the business to predict on.
- *k*: The value of *k* to use for *k*-nearest-neighbors.
- *conf*: The confidence level of the interval (e.g. 95 for a 95% interval)

The function returns a confidence interval for the probability of that business getting a 'Buy' rating (as an array).

Here is the function, partially completed:

```
def prob_interval(train, business, k, conf):
```

```
    probs = _____
           (a)
```

```
    for i in np.arange(10000):
```

```
        boot_data = train._____
                   (b)
```

```
        probs = np.append(probs, _____)
                               (c)
```

```
    left = _____
          (d)
```

```
    right = _____
           (e)
```

```
    return make_array(left, right)
```

i. (2.0 pt) Fill in blank (a).

`make_array()`

ii. (2.0 pt) Fill in blank (b).

`sample()`

iii. (3.0 pt) Fill in blank (c).

```
knn_probs(boot_data, business, k)
```

iv. (3.0 pt) Fill in blank (d).

```
percentile((100-conf)/2, probs)
```

v. (3.0 pt) Fill in blank (e).

```
percentile((100+conf)/2, probs)
```

3. (0.0 points) Optional

(a) Based on the problems in this exam, which show does Prof. Sahai most strongly recommend watching?

☐ The White Lotus

☐ Yellowstone

☒ Succession

☐ Beef

☐ Ted Lasso

(b) If there was any question on the exam that you thought was ambiguous and required clarification to be answerable, please identify the question (including the title of the section, e.g., Experiments) and state your assumptions. Be warned: We only plan to consider this information if we agree that the question was erroneous or ambiguous and we consider your assumption reasonable.



(c) Draw a picture or share a few words describing your experience in Data 8!

